An Phung

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Education

University of British Columbia Expected Graduation: May 2028

Bachelor of Science (B.Sc.) in Computer Science *Vancouver, BC*

Relevant Coursework: Data Structures & Algorithms, Computer Systems, Software Construction, Relational Databases

Awards: Outstanding International Student Award; J Fred Muir Memorial Scholarship in Science; Dean's Scholar (2024-25); Dean's List (2025-26)

Technical Skills

Languages: Java, Python, SQL, JavaScript/TypeScript, C++, Ruby.

Frameworks & Libraries: React, FastAPI, Spring Boot, Apache Kafka, Kafka Streams, Apache Spark, scikit-learn.

Databases: PostgreSQL, MongoDB, Snowflake, Pinecone.

Tools & Platforms: AWS (EC2, S3, EMR, EKS, RDS), GCP, Terraform, Docker, Kubernetes, GitHub Actions, ArgoCD, Datadog, Nginx, Linux.

Technical Experiences

Software Engineering Intern Jan 2026 – Aug 2026

Planview *Vancouver, BC*

- Owned an epic to modernize a legacy **Kafka Streams** metrics platform across 5 services, decommissioning it to free **˜206 GiB RAM** of dedicated infrastructure capacity.
- Consolidated two duplicated **Apache Spark** pipelines on **AWS EMR** into a shared workflow with a common schema, cutting the team's 5 core EMR pipelines' total cost by **15%**.
- Hardened data cleanup across **8 Aurora tables** with a recurring **Spark** garbage-collection job that cross-references **S3** against **Aurora** to detect orphaned records, removing **˜1M+** orphaned records to control RDS storage costs, with auto-rollback and **Datadog** alerting on failure.
- Migrated a service off a shared **Kubernetes** monorepo onto standalone **AWS EKS**, eliminating cascading deployment failures and cutting CI/CD build time from over 1 hour to **8 minutes**.
- Automated **Spark** reprocessing on **AWS EMR** with a custom **Claude Skill**, replacing manual port-forwarding and setup with reusable steps and cutting job setup time from 10 minutes to **1 minute**.

Backend Engineering Intern May 2024 – Aug 2024

Thang Long Software Solution *Hanoi, Vietnam*

- Optimized **PostgreSQL** queries and redesigned schemas for **5,000+ tenants**, reducing query response time by **30%**; traced and resolved backend defects across database and service-layer components.
- Automated backend **fixed-asset depreciation** and tax-reporting workflows in **C++** for a business accounting platform, replacing manual Excel-based reporting across 2 internal departments.

Technical Projects

Fraud Detection – Real-Time ML Pipeline 2026

- Built an event-driven fraud detection pipeline streaming transactions through **Kafka** to a real-time scoring consumer backed by **FastAPI** and **PostgreSQL**.
- Addressed extreme class imbalance (0.17% fraud) with an unsupervised **Isolation Forest** trained on normal-only transactions, avoiding reliance on labeled fraud data.
- Deployed via **GitOps** (**ArgoCD**, **GitHub Actions**) to **Kubernetes** on **Terraform**-provisioned **GCP**, with **Prometheus/Grafana** monitoring.

SafeNest – Safety-First Rental Platform 2026

- Built a full-stack rental platform solo in **Ruby on Rails 8.1** and **Hotwire**, modeling the application lifecycle as a 9-state machine with a whitelisted transition guard.
- Implemented ownership-scoped authorization across 12 controllers and **Stripe** payments with signature-verified webhooks; validated with **158 RSpec** tests.

QuickNote – AI Document Q&A Platform 2025

- Built a full-stack **RAG** platform with **React/FastAPI**, using **JWT** auth and an ingestion pipeline generating **OpenAI** embeddings from **S3** uploads, backed by **PostgreSQL**.
- Added per-user daily **rate limiting** on **LLM** calls to keep API costs predictable and prevent abuse.

Additional Experience

Undergraduate Teaching Assistant Sep 2025 – Dec 2025

University of British Columbia *Vancouver, BC*

- Led weekly Calculus tutorials for 100+ students, clarifying core concepts and grading coursework.